

SARDAR PATEL UNIVERSITY

B.C.A Semester – 6 (Reg. & NC)(CBCS)

Thursday, Date: 30th March 2017

Session: Evening Time : 02:00 P.M. to 5:00 P.M.

Subject Code: US06BCA02

Subject Title : Computer Networks

Total Marks: 70

Q1. Multiple Choice Questions. (Attempt all)

[10]

1. "Interconnected collection of autonomous computers" is known as _____.
A. Computer Architecture C. Computer Network
B. Subnet D. Host
2. When the packets are small and all the same size, they are often called _____.
A. Cells C. Trunks
B. Frames D. Groups
3. All programs, equipment's & specially data can be made available to anyone on a network without regard to the physical location of the resource & user is termed as _____.
A. Reliability C. Increased productivity
B. Reducing Cost. D. Resource Sharing
4. _____ transmission media provides fastest mode of delivery of data signals.
A. Twisted pair C. Magnetic media
B. Copper wire D. Fiber Optics
5. _____ is a phenomenon that occurs then a beam of light travels through media of two different densities, if the angle of incidence is greater than the critical angle.
A. Reflection C. Refraction
B. Transition D. Equilibrium
6. _____ carry signals of higher frequency than _____.
A. Twisted pair ; Fiber optics C. Coaxial cable ; Fiber optics
B. Coaxial cable ; Twisted pair D. None of these
7. Mail and directory service is responsibility of _____ layer.
A. Physical C. Application
B. Session D. Presentation
8. A set of layers and protocols is called _____.
A. Protocol Hierarchy C. Network Architecture
B. Layers D. Protocol Stack
9. _____ networking device can be used to connect two different networks that use different models.
A. Hubs C. Gateway
B. Amplifier D. Repeater
10. A router forwards packets on the basis of their _____ address.
A. Port C. Logical
B. Source D. MAC

Q2. Answer the following short questions (Attempt any TEN) [20]

1. Explain DQDB used in MAN networks with diagram.
2. Explain components of SUBNET with diagram.
3. Define the following terms : i) Topology ii) Data rate
4. Draw figure of Serial transmission and parallel transmission.
5. What is modulation? List three modulation techniques.
6. What does MODEM stand for? Draw figure showing working of MODEM.
7. Define i) Protocol ii) Layer
8. What is footprint?
9. What is orbit?
10. State are the advantages of star topology over mesh topology?
11. Explain how does Broadcast networks differ from point-to point networks?
12. State the main difference between repeater and bridge?

Q3. What is Computer Network? Explain LAN and WAN network categories in detail. [10]

OR

Q3. Explain the advantages of computer networks in detail. [10]

Q4.a. What is transmission media? List different transmission media and explain any one in detail. [06]

b. Write the difference between circuit switching and packet switching. [04]

OR

Q4.a. What do you mean by Multiplexing? Explain any one Multiplexing technique in detail. [06]

b. Explain Synchronous transmission in detail. [04]

Q5.a. What is OSI model? List all layers of OSI model. Explain all functionalities of any one layer of OSI model. [05]

b. Write a note on protocol hierarchies. [05]

OR

Q5.a. Draw diagram showing different layers of OSI model and List main responsibilities of all layers of OSI model with the protocol used at that layer. [05]

b. Write a note on communication satellite. [05]

Q6.a. Explain Mesh and Star LAN topologies in detail. [06]

b. State various forms of CSMA techniques. Explain any one in detail. [04]

OR

Q6.a. Explain Bus and Ring LAN topologies in detail. [06]

b. Explain working of Router and Hubs in brief. [04]